

Unit: Unit 2: Exponents Radicals
Topic: Operations with Radicals

Name: _____

Date: _____

Homework (Carolina Reaper)

Open Ended Questions (FRQ)

Q9. A composer writes a piece of music with a tempo of $3\sqrt{2}$ beats per minute. If the composer wants to increase the tempo by a factor of

$\frac{1}{3}$, what is the new tempo? Show your work and explain your reasoning.

Q10. A sculptor creates a sculpture with a height of $9\sqrt{5}$ inches and a base of $3\sqrt{5}$ inches. If the sculptor wants to create a smaller version with a height that is

$\frac{1}{3}$ times the original height, what is the new base dimension? Show your work and explain your reasoning.

Chef's Tips

- Make sure to check the units of the answers.
- Encourage students to show their work and explain their reasoning.
- Remind students to use the properties of radicals to simplify their answers.